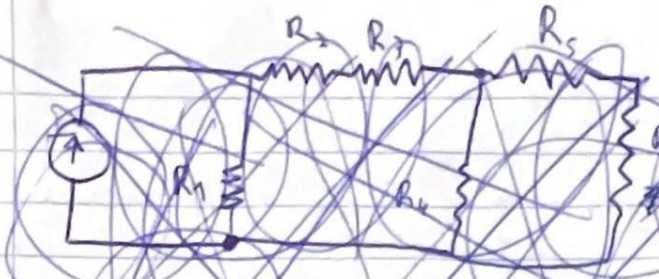
c)



$R_1 = 300\Omega$ $R_4 = 100\Omega$
$R_2 = 50\Omega$ $R_5 = 80\Omega$
$R_3 = 90\Omega$ $R_6 = 70\Omega$

$R_{2,3} + R_5 = 140 + 80 = 220\Omega$

$R_2 + R_3 = 50\Omega + 90 = 140\Omega$

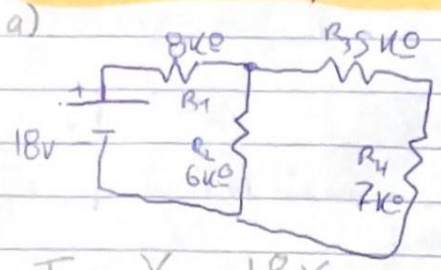
$R_1 // R_4 = \frac{300 \cdot 100}{400} = 75\Omega$

$R_{1,4} // R_6 = \frac{70 \cdot 70}{145} = 35,2\Omega$

$\Rightarrow R_{2,3,5} + R_{1,4,6} = 220 + 36,2\Omega = 256,2\Omega$

3.5)

a)

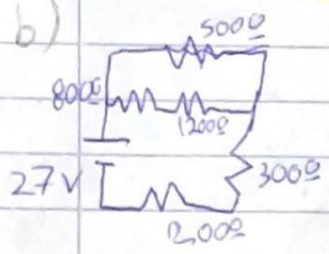


- $R_3 + R_4 = 12\,k\Omega$
- $\frac{R_2 \cdot R_{34}}{R_2 + R_4} = \frac{72k\Omega}{18} = 4k\Omega$

$I = \frac{V}{R} = \frac{18V}{12} = 1,5mA$ $R_{eq} = R_1 + R_{234} = 8 + 4 = 12k\Omega$

$P = 18V \cdot 1,5mA = 27mW$

b)



$R_1 = 500\Omega$
$R_2 = 800\Omega$
$R_3 = 1200\Omega$
$R_4 = 300\Omega$
$R_5 = 200\Omega$

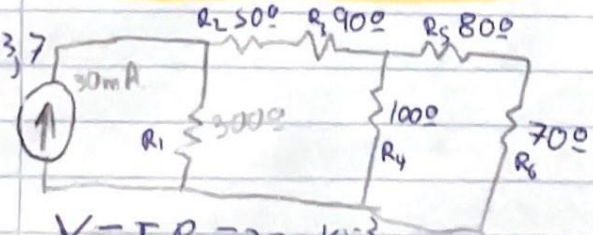
- $R_2 + R_3 = 2000\Omega$
- $\frac{R_{23} \cdot R_1}{R_{23} + R_1} = \frac{2000 \cdot 500}{2500} = 400$

$R_{123} + R_5 + R_4 = 400 + 300 + 200 = 900\Omega$

$P = \frac{V^2}{R_{eq}} = \frac{27^2 V}{900\Omega} = 0,81W$

c)

$P = I \cdot V$
$= 30 \cdot 10^{-3} \cdot 3,7$
$= 0,11W$



$R_6 + R_5 = 150\Omega$

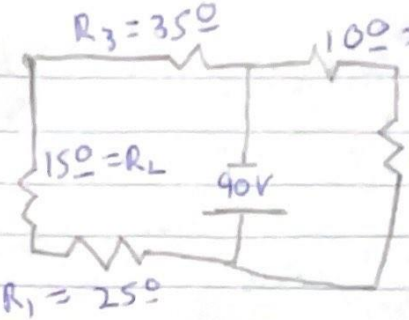
$R_{56} // R_4 = \frac{150 \cdot 100}{250} = 60$

$R_{456} + R_2 + R_3 = 50 + 90 + 60 = 200\Omega$

$R_{eq} = R_1 // R_{...} = \frac{200 \cdot 300}{500} = 120\Omega$

$V = I R = 30 \cdot 10^{-3} \cdot 123,5 = 3,7V$

3.5 δ)



$R_4 + R_5 = 50\Omega$

$R_3 + R_2 + R_1 = 75\Omega$

$R_{123} // R_{45} = \frac{75 \cdot 50}{125}$

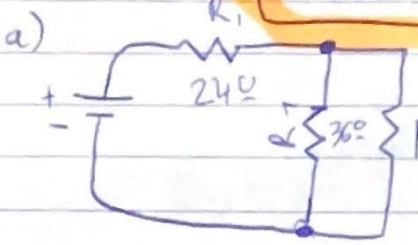
$R_{eq} = 30\Omega$

ii) $V = IR \Rightarrow I = \frac{V}{R}$

$I = \frac{90}{30} = 3\,A$

$P = 90 \cdot 3 = 270\,W$

3.6 a)



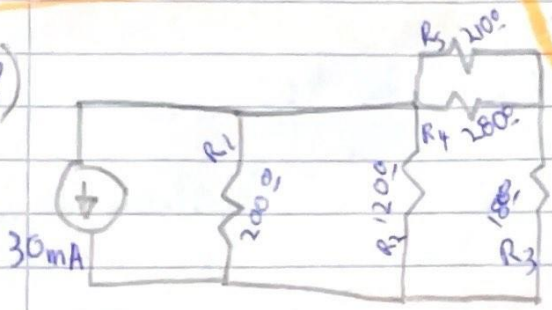
i) $R_{eq} = R_2 // R_3 = \frac{36 \cdot 18}{36+18} = 12\Omega$

$R_{eq} = R_1 + R_{eq_1} = 12 + 24\Omega = 36\Omega$

ii) $V = IR \Rightarrow I = \frac{V}{R} = \frac{18}{36} = 0,5\,V$

$P = V \cdot I = 0,5 \cdot 18 = 9\,W$

β)



i) $R_4 // R_5 = \frac{210 \cdot 280}{490} = 120\Omega$

$R_{45} + R_3 = 120 + 180 = 300\Omega$

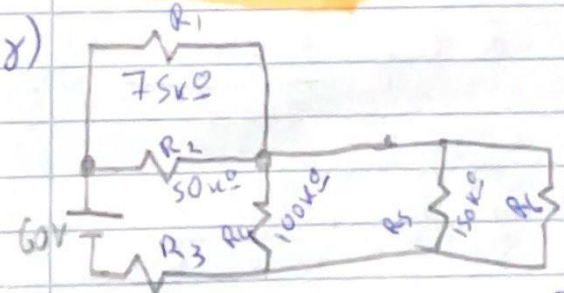
$R_{453} // R_2 = \frac{300 \cdot 120}{420} = 85,7\Omega$

ii) $V = I R_{eq} = 30 \cdot 10^{-3} \cdot 60\Omega = -7,8\,V$

$R_{eq} = R_1 // R_{4532} = \frac{85,7 \cdot 200}{285,7} = 60\Omega$

$P = V \cdot I = -7,8\,V \cdot 30 \cdot 10^{-3}\,A = 54\,mW$

γ)



$R_5 // R_6 = \frac{60 \cdot 50}{210} = 42,85\,k\Omega$

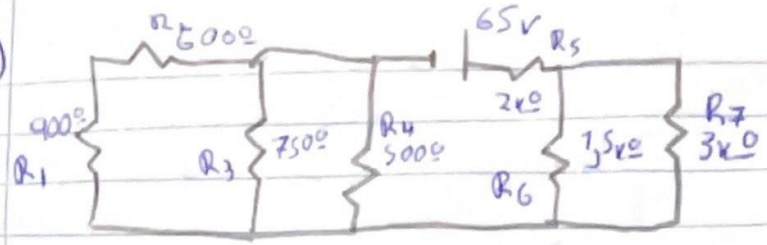
$R_1 // R_2 = \frac{75 \cdot 50}{125} = 30\,k\Omega$

$R_4 // R_{56} = \frac{42,85 \cdot 100}{142,85} = 30\,k\Omega$

$R_{eq} = R_{456} + R_{12} + R_3 = 240\,k\Omega$

ii) $V = I \cdot R \Rightarrow I = \frac{V}{R} = \frac{60}{240} = 0,25\,mA$ / $P = 0,25 \cdot 60 = 15\,mW$

δ)



$R_6 // R_7 = \frac{3 \cdot 1,5}{4,5} = 1k\Omega$

$R_1 + R_2 = 1500\Omega$

$R_{12} // R_3 = \frac{1500 \cdot 750}{2250} = 500\Omega$

$R_{eq} = R_{7234} + R_{67} + R_5$

$= 250\Omega + 1k\Omega + 2k\Omega$

$R_{eq} = 3,25\ k\Omega$

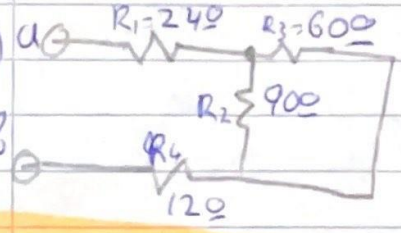
$\Leftarrow R_4 // R_{123} = \frac{500 \cdot 500}{1000} = 250\Omega$

ii) $V = I \cdot R \Rightarrow I = \frac{V}{R} = \frac{65V}{3,25} = 20 \cdot 10^{-3} A$

$P = V \cdot I = 65 \cdot 20 = 1300 \cdot 10^{-3} W$

$= 1,3 W$

3.8) a)

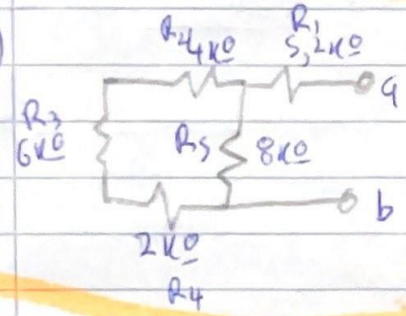


$R_3 // R_2 = \frac{60 \cdot 90}{150} = 36\Omega$

$R_{eq} = R_1 + R_{23} + R_4 = 24 + 36 + 12 = 72\Omega$

$R_{ab} = 72\Omega$

β)



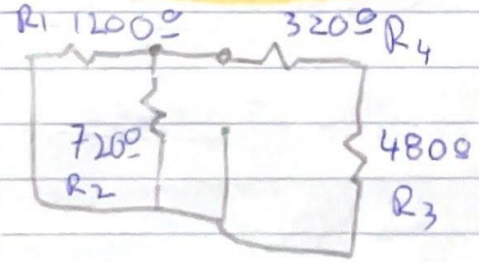
$R_4 + R_3 + R_2 = 2 + 6 + 4 = 12k\Omega$

$R_5 // R_{432} = \frac{12 \cdot 8}{20} = 4,8k\Omega$

$R_{eq} = R_1 + R_{5432} = 4,8 + 5,2$

$R_{ab} = 10\ k\Omega$

γ)



$R_2 // R_1 = \frac{720 \cdot 1200}{1920} = 450\Omega$

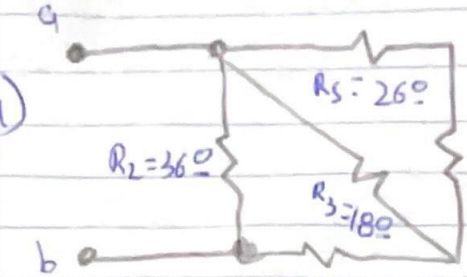
$R_4 + R_3 = 800\Omega$

$R_{ab} = R_{43} // R_{21} = \frac{800 \cdot 450}{1250}$

$= 288\Omega$

3, a)

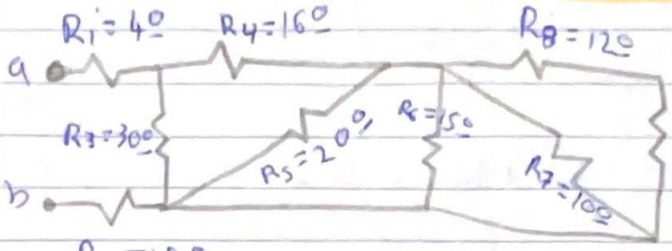
a)



- $R_5 + R_4 = 36\Omega$
- $R_3 // R_{45} = \frac{18 \cdot 36}{54} = 12\Omega$
- $R_{345} + R_1 = 18\Omega = 12 + 6$

$R_{ab} = \cdot R_{7345} // R_2 = \frac{18 \cdot 36}{54} = 12\Omega$

b)



- $R_8 + R_9 = 30\Omega$
- $R_7 // R_{89} = \frac{10 \cdot 30}{40} = 7,5$

$R_6 // R_{789} = \frac{7,5 \cdot 15}{22,5} = 5\Omega$

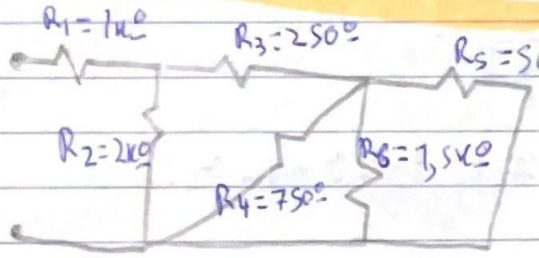
$R_{6789} // R_5 = \frac{20 \cdot 5}{25} = 4\Omega$

$R_{56789} + R_4 = 4 + 16 = 20\Omega$

$R_{456789} // R_3 = \frac{20 \cdot 30}{50} = 12\Omega$

$R_{ab} = R_{3456789} + R_1 + R_2$
$= 12\Omega + 14 + 4$
$= 20\Omega$

γ)



$R_6 // R_5 = \frac{500 \cdot 1500}{2000} = 375\Omega$

$R_{56} // R_4 = \frac{750 \cdot 375}{1125} = 250\Omega$

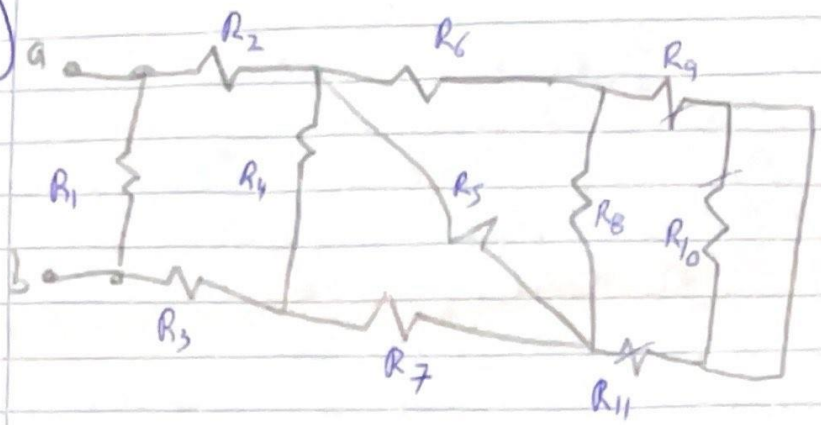
$R_3 + R_{456} = 500\Omega$

$R_2 // R_{3456} = \frac{500 \cdot 2000}{2500} = 400\Omega$

$R_{ab} = R_{23456} + R_1$
$= 7,4\ k\Omega$

$R_1 = 50\Omega$, $R_2 = 25\Omega$, $R_3 = 10\Omega$, $R_4 = 24\Omega$, $R_5 = 40\Omega$, $R_6 = 28\Omega$, $R_7 = 20\Omega$, $R_8 = 16\Omega$, $R_9 = 30\Omega$, $R_{10} = 60\Omega$, $R_{11} = 18\Omega$

d)



- $R_9 + R_{10} + R_{11} = 18 + 60 + 30 = 108\Omega$
- $Req_1 = R_{9,10,11} // R_8 = \frac{16 \cdot 108}{124} = 13,9\Omega$

$Req_2 = Req_1 + R_6 = 13,9 + 28 = 41,9$

$Req_3 = Req_2 // R_5 = \frac{41,9 \cdot 40}{81,9} = 20,46\Omega$

$Req_4 = Req_3 + R_7 = 20,46 + 20\Omega = 40,46\Omega$

$Req_5 = Req_4 // R_4 = \frac{40,46 \cdot 24}{64,46} = 15$

$Req_6 = Req_5 + R_2 + R_3 = 15 + 25 + 10 = 50\Omega$

$Rab = Req // R_1 = \frac{50 \cdot 50}{100} = 25\Omega$ $\boxed{Rab = 25\Omega}$